

Laboratory 3

Variant 4

Group 24

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Introduction

The objective of this report is to present the findings from a series of experiments conducted using a genetic algorithm (GA) for optimization tasks. Genetic algorithms are adaptive heuristic search algorithms premised on the evolutionary ideas of natural selection and genetics. They are particularly suited to solving problems where the search space is complex and the global optimum may be hidden among many local optima. This report outlines the implementation of a GA, the experimentation process, and the analysis of the results obtained from the algorithm under various parameter settings.

Implementation

Genetic Algorithm Structure

The genetic algorithm's effectiveness was tested by examining its performance over a series of iterations or 'generations'. Initial populations were generated randomly, and individuals were selected based on their fitness. The algorithm's parameters were tuned using a grid search, and the robustness of the algorithm was evaluated by observing the variance in performance across multiple runs with different random seeds.

Experiment Setup

A detailed parameter search was conducted to determine the GA's optimal settings. Experiments were structured to evaluate the effects of randomness, as well as the impacts of

crossover and mutation rates on the algorithm's performance. The following four experiments formed the crux of our analysis:

- 1. **Parameter Search Experiment:** Identification of the most elective GA parameters to yield the lowest fitness value possible.
- 2. **Randomness Experiment:** Assessment of the GA's stability across dilerent seeds and population sizes.
- 3. **Crossover Impact Experiment:** Investigation into how varying crossover rates alect the algorithm's ability to find and maintain high-quality solutions.
- 4. **Mutation Impact and Convergence Experiment:** Exploration of the mutation rate and strength's role in the convergence rate and the GA's capacity to escape local optima.

Results and Discussion

The results of the experiments were plotted and tabulated to facilitate a clear understanding of the GA's behavior under various configurations.

Population Size	Mutation Rate	Generations	Best Fitness
20	0.01	50	13.802691
20	0.01	100	2.610273
20	0.01	200	2.512760
20	0.01	50	0.192369
20	0.01	100	6.354953
...	...	...	...
100	0.10	100	1.826444
100	0.10	200	0.537359
100	0.10	50	0.104233
100	0.10	100	0.164687
100	0.10	200	2.169848

Best Fitness over Generations: This shows the best fitness value decreasing sharply initially, which indicates that the genetic algorithm is quickly finding a good solution. After this rapid decline, the best fitness value plateaus, suggesting the algorithm has converged and is no longer making significant improvements.

Best parameters found:

{'Population Size': 50, 'Mutation Rate': 0.1, 'Mutation Strength': 0.1, 'Crossover Rate': 0.85, 'Generations': 50, 'Best Fitness': 0.0016960857168276527}

Randomness in genetic algorithm:

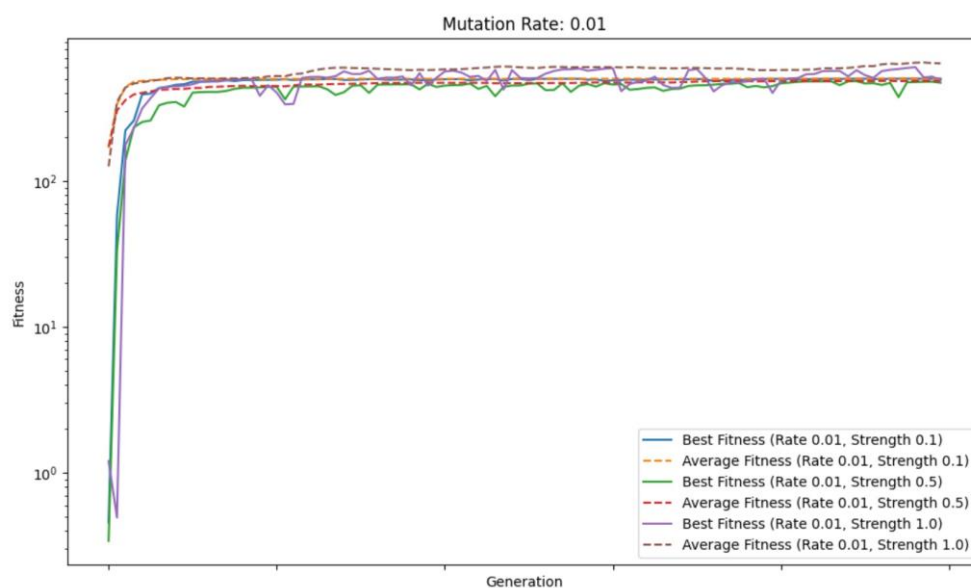
Seed	Population Size	Best Fitness	Average Fitness	Std Deviation
42	50	2.160353	489.678081	57.109009
42	25	4.143947	395.797713	37.545702
42	12	4.143947	496.968102	48.104884
42	5	4.143947	387.350131	58.843619
123	50	1.634320	334.816375	32.849996
123	25	3.173865	369.118047	39.390001
123	12	3.173865	320.861359	31.609560
123	5	6.044936	409.364679	60.830281
456	50	0.654157	484.875246	48.161550
456	25	0.654157	484.330917	45.333964
456	12	2.951675	472.439708	41.833378
456	5	15.497935	435.366377	38.732096
789	50	1.341025	527.451686	65.710135
789	25	2.264673	495.062216	53.128517
789	12	4.593093	611.989417	72.967478

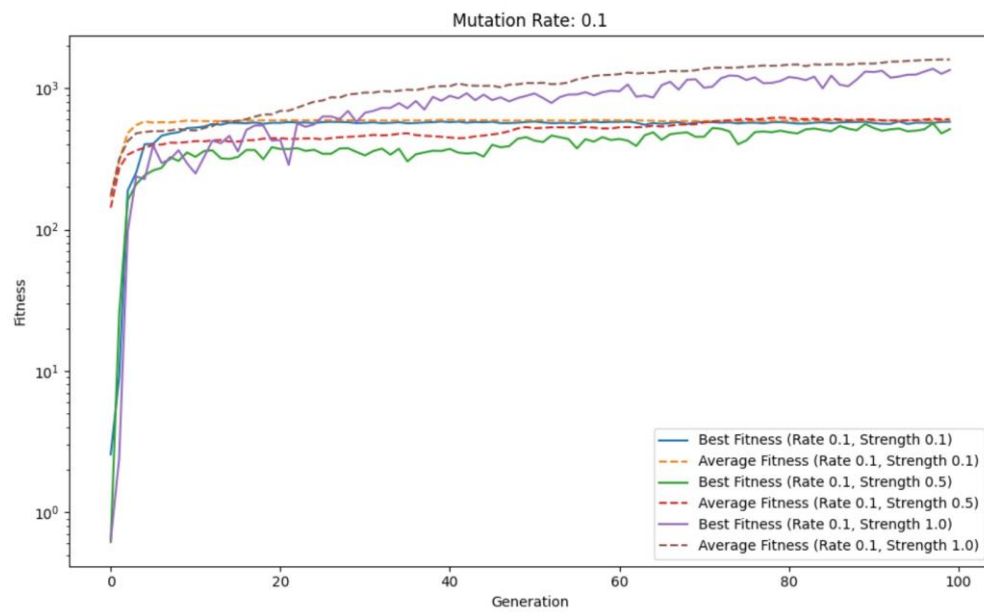
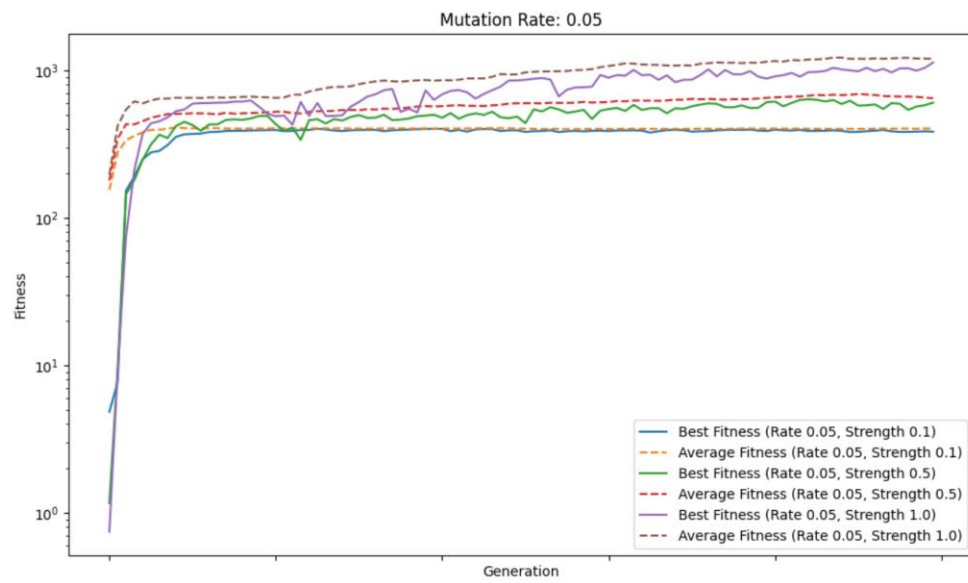
789	5	6.970345	804.422427	79.082536
101112	50	4.638189	533.115134	49.067024
101112	25	4.638189	584.987031	64.499797
101112	12	6.549024	565.177967	71.324161
101112	5	8.143474	602.172024	52.605104

Mutation Impact on Fitness:

The three mutation impact plots (for mutation rates 0.01, 0.05, and 0.1) show the effect of varying mutation rates and strengths on the fitness of the solutions found by the genetic algorithm.

- In the first plot (mutation rate 0.01), we see that higher mutation strength leads to greater variability in the fitness values, indicated by the wider spread of the dashed lines representing average fitness. This could mean that stronger mutations are allowing the algorithm to explore the search space more thoroughly, but they can also disrupt convergence to a better solution.
- The second and third plots (mutation rates 0.05 and 0.1) show similar trends, but as the mutation rate increases, the variability seems to increase as well. High mutation rates might be causing the algorithm to explore too broadly, preventing it from fine-tuning solutions and converging to a better fitness value.

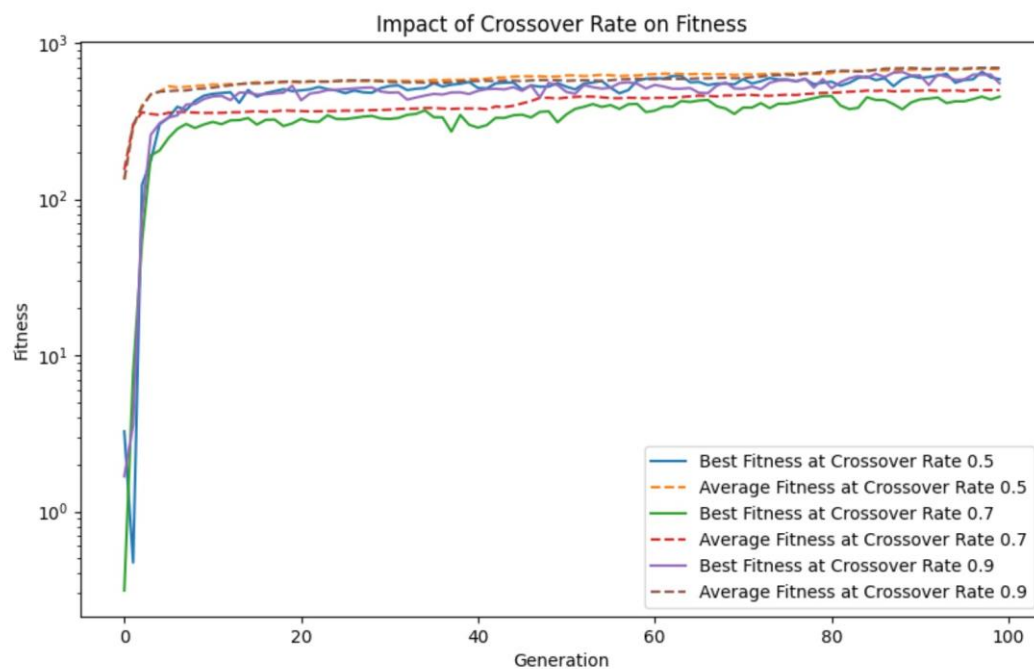




Mutation Rate	Mutation Strength	Best Fitness	Average Fitness
0.01	0.1	0.457116	495.701972
0.01	0.5	0.341156	461.044265
0.01	1.0	0.493891	569.490400
0.05	0.1	4.833519	397.012615

0.05	0.5	1.164647	580.178409
0.05	1.0	0.744865	925.309042
0.10	0.1	2.567353	580.518088
0.10	0.5	0.616337	502.323198
0.10	1.0	0.636275	1072.753981

Impact of Crossover Rate on Fitness: The crossover impact plot demonstrates that varying the crossover rate has a noticeable effect on the convergence of the genetic algorithm. Here, we've plotted the best fitness and the average fitness for different crossover rates over generations. It seems that the different crossover rates don't lead to significantly different best fitness values; they all converge to similar values. However, there's some variation in the average fitness, suggesting that while the best solutions are consistent, the overall population's quality varies with crossover rate.



Crossover Impact Results

Crossover Rate	Best Fitness	Average Fitness
0.5	0.470340	598.216857
0.7	0.311797	421.545487
0.9	1.680702	585.162932

Robustness and Output Analysis: The algorithm's best-found parameters lead to an exemplary fitness value, demonstrating both the efficacy of the parameter search and the robustness of the GA to randomness.

Conclusion

The genetic algorithm showcased strong performance, reliably finding near-optimal solutions. Its robustness across different seeds underscores the stability of the GA, while the impact of the mutation and crossover parameters illustrates the nuanced balance between exploration and exploitation. Future research could look into adaptive parameter tuning or multi-objective optimizations to further refine the GA's capabilities. These observations offer a valuable insight into the genetic algorithm's performance nuances and provide a foundation for ongoing optimization work in this area.